Final-Exam Review

CS 340

About Final-Term Exam

- The exam starts on Wednesday, 27 April 2022, 9:00 AM, and will close on Wednesday, 27 April 2022, 11:15 AM.

- Time: 2 hours

- The exam has 22 Questions including Multiple-choice, True/False, Matching, and one needs a written answer.

- In some questions, more than one choice can be true and you need to choose all correct answers.

- Proctortrack Onboarding

Examples

- Be able to find the complexity class given any function
- Be able to compare two functions regarding their complexities
- Be able to perform best-, worst-, and average-case analysis given an algorithm
- Understand how a specific algorithm work

Exercise 5 : Algorithm Design Techniques [0.5x5+0.5x5=5pts]

When introducing the different algorithm design techniques we focused on the following five common types of algorithms :

Greedy algorithm, Divide and conquer, Dynamic Programming, Randomized Algorithms, Backtrack Search Algorithms.

1. To which of the above types belongs each of the following algorithms :

   (a) Huffman Coding
   (b) Prim's Algorithm
   (c) Quicksort
   (d) Mergesort
   (e) Kruskal's Algorithm

2. Which design technique can be used to solve each of the following problems :

   (a) The loading problem
   (b) The 0/1 Knapsack problem
   (c) Ordering Matrix Multiplications
   (d) Closest-Points Problem in $O(N \log N)$

Exercise 1 : Algorithm Analysis [6pts]

1. Give an analysis of the running time (Big-Oh notation) for each of the following 4 program fragments. Note that the running time corresponds here to the number of times the operation `sum++` is executed. `sqrt` is the function that returns the square root of a given number.

(b)
```
sum = 0;
for(i=0;i<sqrt(n)/2;i++)
  for(j=i;8+i;j++)
   for(k=j;k<8+j;k++)
    sum++;
```

(c)
```
sum = 0;
for(i=1;i<2*n;i++)
 for(j=1;j<i*i;j++)
   for(k=1;k<j;k++)
     if (j % i == 1)
       sum++;
```

(d)
```
sum = 0;
for(i=1;i<2*n;i++)
 for(j=1;j<i*i;j++)
   for(k=1;k<j;k++)
     if (j % i)
       sum++;
```

(b) $O(\sqrt{N})$, **(c)** $O(N^4)$ **and (d)** $O(N^5)$

(a)
```
sum = 0;
for(i=0;i<n;i++)
 sum++;
for(j=0;j<n;j++)
 sum++;
```

(b)
```
sum = 0;
for(i=0;i<n/2;i++)
  for(j=i;j<n/4;j++)
    sum++;
```

Solution

(a) $O(N)$.

(b) $O(N^2)$.

Example

- We assume we have N naturals to sort. Determine the correct running time (big-Oh notation) of each of the following four algorithms: Insertion sort, Heapsort, Mergesort and Quicksort, in the case of **a randomly generated input.**

- Heapsort runs in? O(N log N)
- Mergesort runs in? O(N log N)

Quick Sort!

- Suppose that the splits at every level of quicksort are in the proportion 1 - α to α, where $0 < \alpha \leq 1/2$ is a constant. Show that the minimum depth of a leaf in the recursion tree is approximately - lg n/lg α and the maximum depth is approximately - lg n/ lg(1 - α). (Don't worry about integer round-off.)

Proof

The minimum depth of the tree is the solution of $n\alpha^x \le 1$:

$$n\alpha^x \le 1$$
$$\Downarrow$$
$$\alpha^x \le \frac{1}{n}$$
$$\Downarrow$$
$$x \ge \log_\alpha \frac{1}{n}$$
$$\Downarrow$$
$$\log_\alpha \frac{1}{n} = \log_{1/\alpha} = \frac{\lg n}{\lg(1/\alpha)} = -\frac{\lg n}{\lg \alpha}$$

In the same way, the maximum depth is $\log_{1/(1-\alpha)} n = -\frac{\lg n}{\lg(1-\alpha)}$

Application

- Suppose that the splits at every level of quicksort algorithm are in the proportion 5/100 to 95/100.

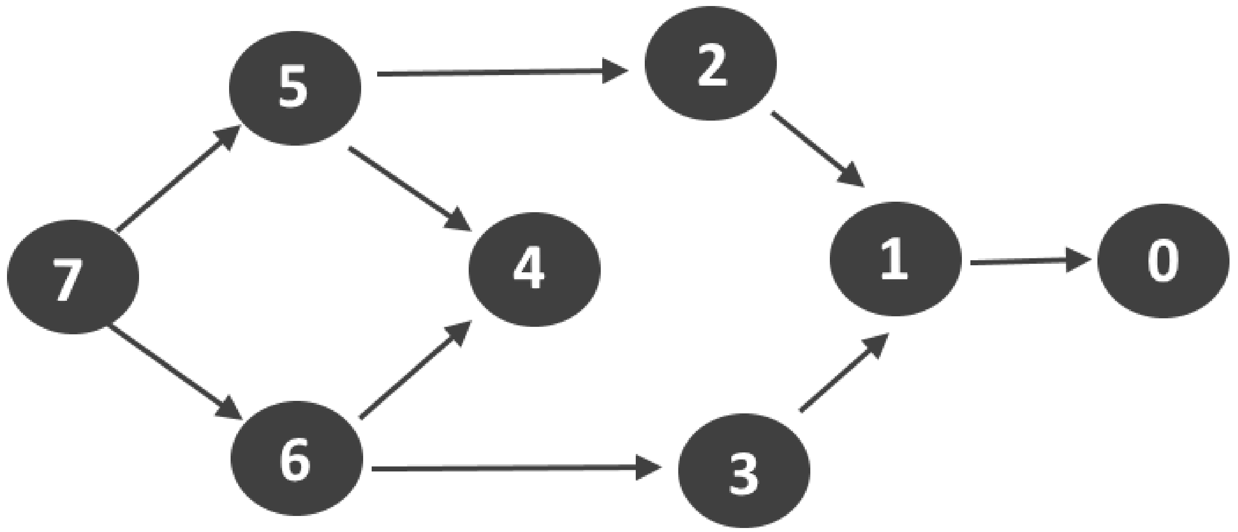
- The maximum depth of a leaf in the recursion tree?

  log 100/95 N

- The minimum depth of a leaf in the recursion tree?

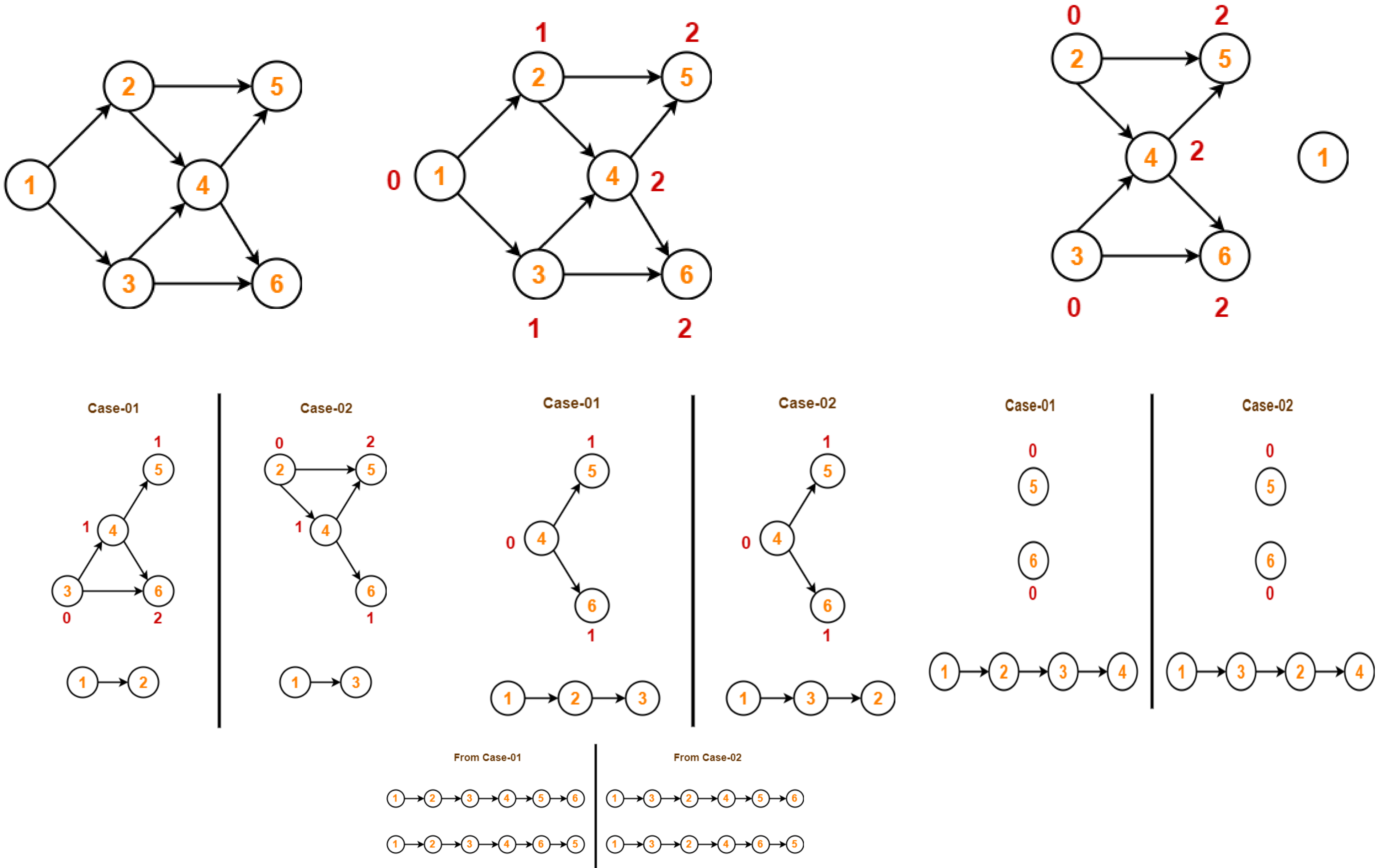
Topological sort

Consider the graph listed below



A possible topological sort is?

Topological Sort : 7 6 5 4 3 2 1 0



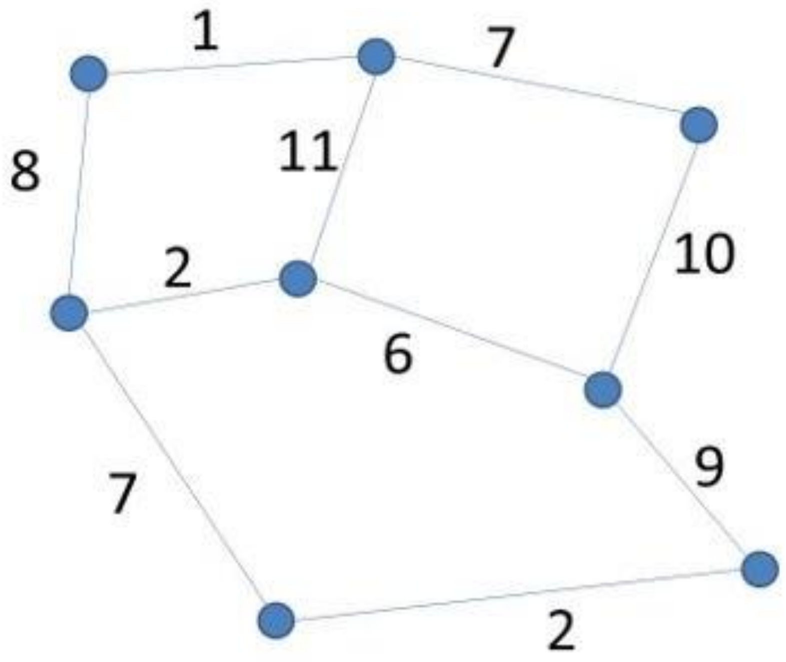
Case-01
Case-02
Case-01
Case-02
Case-01
Case-02
From Case-01
From Case-02

Minimum Spanning Tree

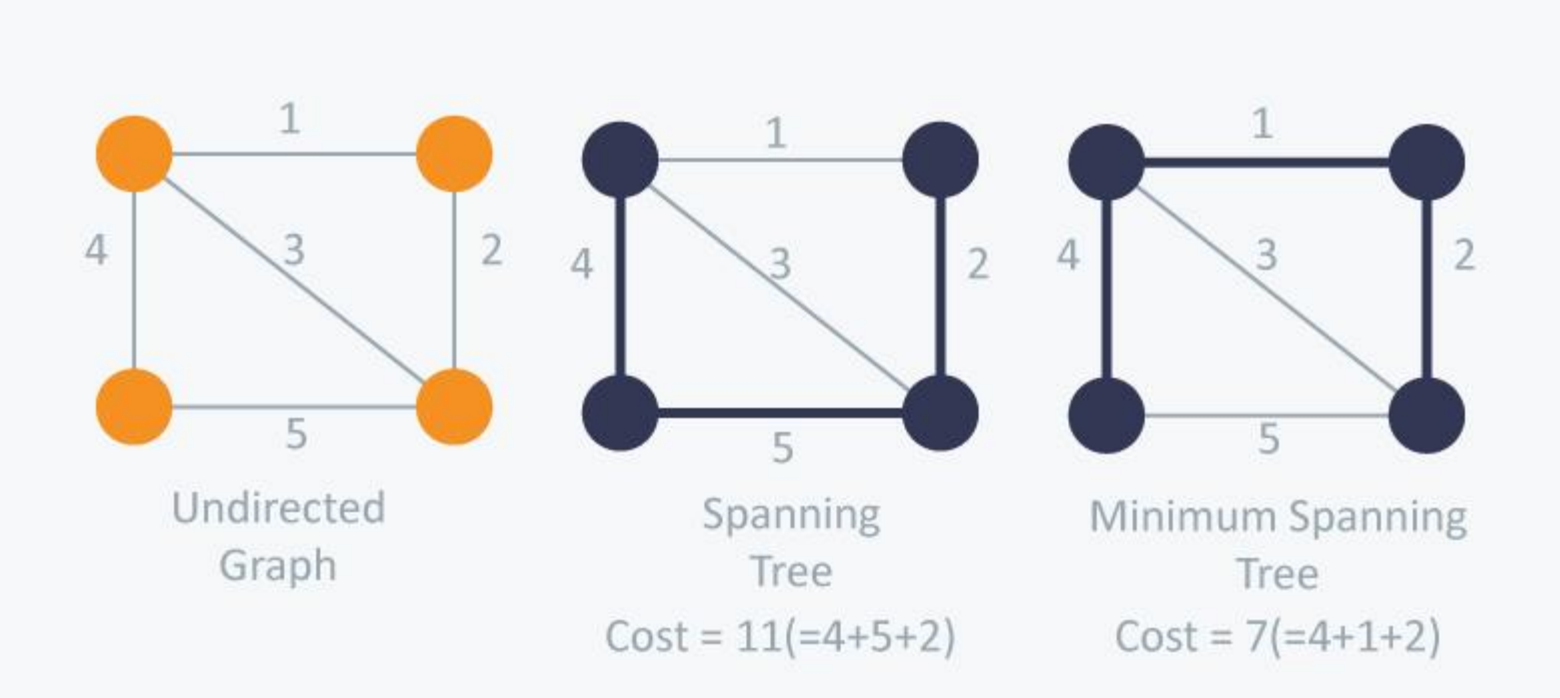
- Let us consider the following weighted graph.

- There exists more than one minimum spanning tree in this graph.
- The minimum spanning tree does not exist .
- A minimum spanning tree exists and has a total weight of X
- A minimum spanning tree exists and has a total weight more than Y
- A minimum spanning tree exists and has a total weight of Z

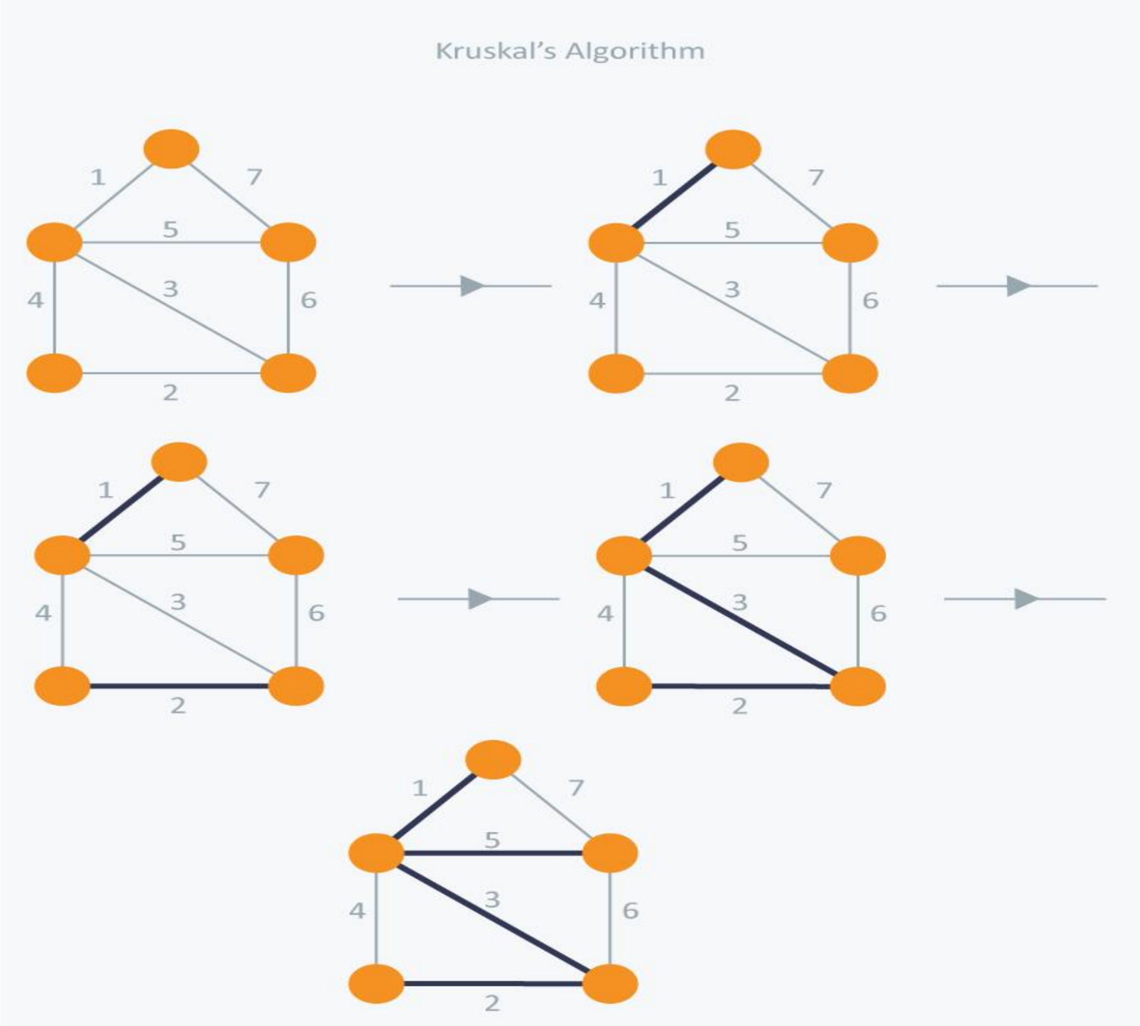
Minimum Spanning Tree



Minimum Spanning Tree



Minimum Spanning Tree



Others

- One from AVL tree
- TSP